

Cryptology Assignment - 3

classmate

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- ① Determine the Multiplicative Inverse modulo 26 for each of the following integers.
1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25,
mod 26

→ $3^{-1} \text{ mod } 26$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1) \rightarrow \textcircled{1}$$

$$2 = 26 + 3(-8) \rightarrow \textcircled{2}$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$\underline{\underline{3^{-1} = 9}}$$

$5^{-1} \text{ mod } 26$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\underline{\underline{5^{-1} = 21}}$$

$7^{-1} \text{ mod } 26$

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\underline{\underline{7^{-1} = 15}}$$

$9^{-1} \text{ mod } 26$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)] \cdot [1] = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\underline{\underline{9^{-1} = 3}}$$

$11^{-1} \text{ mod } 26$

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + \cancel{11} \cdot (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\underline{\underline{11^{-1} = 19}}$$

$15^{-1} \text{ mod } 26$

$$26 = 15(1) + 11$$

$$11 = 26 + 15(-1)$$

$$15 = 11(1) + 4$$

$$4 = 15 + 11(-1)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2))(-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + (26 + 15(-1))(-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$\underline{15^{-1} = 7}$$

$$\underline{17^{-1} \bmod 26}$$

$$26 = 17(1) + 9 \quad 9 = 26 + 17(-1)$$

$$17 = 9(1) + 8$$

$$8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$\underline{17^{-1} = 23}$$

$$\underline{19^{-1} \bmod 26}$$

$$26 = 19(1) + 7$$

$$7 = 26 + 19(-1)$$

$$19 = 7(2) + 5$$

$$5 = 19 + 7(-2)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + (7 + 5(-1))(-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + 7(-4) + 5(2) = 1$$

$$19 + (26 + 19(-1))(-4) + (19 + 7(-2))(2) = 1$$

$$19 + \cancel{26} \cdot 26(-4) + 19(4) + 19(2) + 7(-4) = 1$$

$$19(11) + 26(-8) = 1$$

$$\underline{19^{-1} = 11}$$

$$\underline{21^{-1} \bmod 26}$$

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$21 = 5(4) + 1$$

$$1 = 21 + 5(-4)$$

$$21 + 5(-4) = 1$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21^{-1} = 5$$

$$\underline{23^{-1} \text{ mod } 26}$$

$$26 = 23(1) + 3$$

$$3 = 26 + 23(-1)$$

$$23 = 3(7) + 2$$

$$2 = 23 + 3(-7)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + (23 + 3(-7))(-1) = 1$$

$$26 + 23(-1) + 23(-1) + (26 + 23(-1))(7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$\underline{23^{-1} = 17}$$

$$\underline{25^{-1} \text{ mod } 26}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\underline{25^{-1} = 25}$$

② Choose Any String of your choice to encrypt using the Affine cipher with key values.
multiplier: 5, Additive: 8

$$\rightarrow k_1 = 5, k_2 = 8.$$

$$C = (k_1 P + k_2) \text{ mod } 26$$

H E L L O

7 4 11 11 14

$$H = (5 \times 7 + 8) \text{ mod } 26$$

$$43 \text{ mod } 26 = 17 \rightarrow R$$

$$C = (5 \times 4 + 8) \bmod 26$$

$$28 \bmod 26 = 2 \rightarrow C$$

$$L = (5 \times 11 + 8) \bmod 26$$

$$63 \bmod 26 = 11 \rightarrow L$$

$$L = 11$$

$$O = (5 \times 14 + 8) \bmod 26$$

$$78 \bmod 26 = 0 \rightarrow A$$

HELLO $\rightarrow$ RCLLA

③ Encrypt "HI" using Affine cipher with
 $a = 11$ & $b = 5$

$\rightarrow$ gn. $a = 11$, $b = 5$, plaintext = HI
 $H = 7$, $I = 8$

$$C = (P \times a + b) \bmod 26$$

$$\underline{H = 7}$$

$$\begin{aligned} C &= (11 \times 7 + 5) \bmod 26 \\ &= (77 + 5) \bmod 26 \\ &= 82 \bmod 26 \\ &= \underline{4} \end{aligned}$$

$$\underline{4 = E}$$

$$\underline{I = 8}$$

$$\begin{aligned} C &= (11 \times 8 + 5) \bmod 26 \\ &= (88 + 5) \bmod 26 \\ &= 93 \bmod 26 \\ &= \underline{15} \end{aligned}$$

$$\underline{15 = P}$$

ciphertext = EP

④ The ciphertext UCR was encrypted using affine function $9x + 2 \bmod 26$ find plaintext.

→ ciphertext: UCR

- Affine Encryption: $C = (aP + b) \bmod 26$

- ~~affine~~ 26

$$U = 20, C = 2, R = 17$$

$$C = (aP + b) \bmod 26$$

$$P = a^{-1}(C - b) \bmod 26$$

$$a = 9$$

$$b = 2$$

a^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

$$U = 3(20 - 2) \bmod 26 = 3(18) \bmod 26 = 54 \bmod 26 = 2 = C$$

$$C = 3(2 - 2) \bmod 26 = 0 = A$$

$$R = 3(17 - 2) \bmod 26 = 3(15) \bmod 26 = 45 \bmod 26 = 19 \rightarrow T$$

UCR → CAT

⑤ Decrypt the cipher text using affine cipher, KD YXJUVWDEXZY using the key 3 (multiplier) + 4.

→ $a = 3, b = 4$

Encryption: $C = (aP + b) \bmod 26$

Decryption: $P = a^{-1}(C - b) \bmod 26$

• $3^{-1} \bmod 26$

$$26 = 3(8) + 2 \quad \cdot \quad 2 = 26 + 3(-8)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + (26 + 3(-8))(-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9 = a^{-1}$$

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$$k = 9(10-4) \bmod 26 = 9(6) \bmod 26 = 2 = G$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$Y = 9(24-4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$J = 9(9-4) \bmod 26 = 9(5) \bmod 26 = 19 = T$$

$$U = 9(26-4) \bmod 26 = 9(22) \bmod 26 = 14 = O$$

$$W = 9(22-4) \bmod 26 = 9(18) \bmod 26 = 6 = b$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$E = 9(4-4) \bmod 26 = 9(0) \bmod 26 = 0 = A$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$Z = 9(25-4) \bmod 26 = 9(21) \bmod 26 = 7 = H$$

$$Y = 9(24-4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

Plaintext: CRYPTOGRAPHY.

- ⑥ Use $k_1 = 13$ (Multi), $k_2 = 4$ (even though it is not valid key. try to Encrypt. Encrypt the bring 'input' & 'cipher' above key & find what is true.

$$\rightarrow C = (P \times \text{Multi} + \text{Add}) \bmod 26$$

$$C = (P \times a + b) \bmod 26$$

$$a = 13, b = 4.$$

13 is not coprime of 26.

So we can't find inverse of 13 mod 26

$\therefore$ it is not valid key for affine cipher.

for 'input'

$$I = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (2 \times 13 + 4) \bmod 26 = 30 \bmod 26 = 4 = E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

ERROR

for 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = K$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = K$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R$$

ERRER

Both plaintext will produce the same ciphertext.
The Encryption is not one is one & decryption
is not possible as 13 is not valid key.

⑦ Find the inverse matrix for the key. $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26.$$

$$\det(K) = 3 \times 7 - 2 \times 5 = 21 - 10 = \underline{11}$$

11⁻¹

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(1) = 1$$

$$26 + 11(-2) + \cancel{11}(-1) + 4(-2)(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 16(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = \underline{19}$$

Adjoint of matrix:

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 5 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

⑧ find the inverse of the following matrix whose entries are considered

module 26 : $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

→ $K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \text{ mod } 26$

$\det[K] = 11 \times 3 - 13 \times 2$
 $= 33 - 26 = 7$

$26 = 7(3) + 5$ $5 = 26 + 7(-3)$

$7 = 5(1) + 2$ $2 = 7 + 5(-1)$

$5 = 2(2) + 1$

$5 + 2(-2) = 1$

$26 + 7(-3) + 2(7 + 5(-1))(-2) = 1$

$26 + 7(-3) + 7(-2) + 5(2) = 1$

$26 + 7(-5) + (26 + 7(-3))(2) = 1$

$26 + 7(-5) + 26(2) + 7(-6) = 1$

$26(3) + 7(-11) = 1$

$26 - 11 = 15$

Adjoint is

$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$

$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \text{ mod } 26$

$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26$

$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$

9) Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

- for a key to be valid Δ to determinant should be coprime Δ shouldn't be square.

$$\det(k) = 2(21) - 14(13)$$

$$= 42 - 182$$

$$= -140$$

$$-140 \bmod 26 = 16$$

$$\det k = 16 \bmod 26$$

16 is not coprime to 26

$\therefore$ it will not have an inverse mod 26

$\therefore$ The key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid.

10) Encrypt using 2×2 Hill cipher for plaintext.
Solved using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 = 3 \text{ which is coprime.}$$

SO | LV | ED

$$C = K \times P$$

$$P = \begin{bmatrix} 18 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 = 10 = K$$

$$280 \bmod 26 = 20 = U$$

$$L \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26$$

$$201 \bmod 26 = 19 = T$$

$$308 \bmod 26 = 22 = W$$

$$E \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} \bmod 26 = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26$$

$$36 \bmod 26 = 10 = K$$

$$61 \bmod 26 = 9 = J$$

SOLVED $\rightarrow$ KUTWKJ